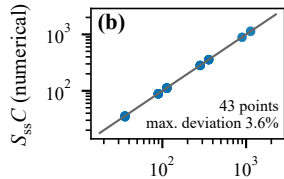
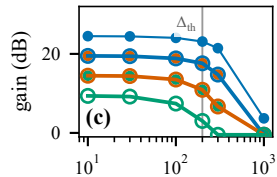


measurement time (ms)

$\bar{n} = 10^6$      $\bar{n} = 10^9$   
 $\bar{n} = 10^7$      $\bar{n} = 10^{10}$   
 $\bar{n} = 10^8$      $\square$  twisting, Lorentzian to Voigt

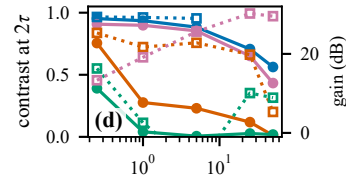


$4g\sqrt{\eta_d\bar{n}}/\kappa$



$\Delta/2\pi$  (MHz)

$\bigcirc$   $N=10^9, \bar{n}=10^8$      $\bullet$   $N=10^{10}, \bar{n}=10^9$   
 $\bullet$   $N=10^9, \bar{n}=10^9$      $\bigcirc$   $N=10^{11}, \bar{n}=10^8$   
 $\bigcirc$   $N=10^{10}, \bar{n}=10^8$      $\bullet$   $N=10^{11}, \bar{n}=10^9$



echo half-time  $\tau$  (ms)

$\bullet$   $\Delta/2\pi = 30$  MHz     $\bullet$   $\Delta/2\pi = 300$  MHz  
 $\bullet$   $\Delta/2\pi = 200$  MHz     $\bullet$   $\Delta/2\pi = 1000$  MHz